Subject: OPTIMIZATION TECHNIQUES

Subject Code: 23AS2402e

UNIT II : Transportation and Assignment

Syllabus: Transportation model, initial basic feasible solutions, optimality (MODI), unbalanced problems, maximization, degeneracy; assignment problem, Hungarian algorithm, multiple solutions, unbalanced and maximization cases.

I. Transportation Problem – Introduction

Definition:

The transportation problem is a special LPP that finds the least-cost plan to ship a single commodity from m sources (origins, supply a_i) to n destinations (demand b_j).

- Cost c_{ij} = cost of shipping one unit from source i to destination j. x_{ij} = units shipped.

Mathematical model

Formula:

Min $Z = \Sigma \Sigma c_{ij} x_{ij}$ $(i = 1..m, j = 1..n)$

$\Sigma_j x_{ij} = a_i$ (supply), $\Sigma_i x_{ij} = b_j$ (demand), $x_{ij} \geq 0$

- **Balanced** problem: $\Sigma a_i = \Sigma b_j$. Unbalanced otherwise.
- A balanced problem always has a feasible solution; a basic feasible solution has exactly $m + n - 1$ allocations (occupied cells).

Transportation table (running example)

From / To	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9

S3	40	8	70	20	18
Demand	5	8	7	14	34

Note:

Total supply = 7 + 9 + 18 = 34 = total demand 5 + 8 + 7 + 14 → balanced. m + n − 1 = 3 + 4 − 1 = 6 allocations are needed.

Steps of the solution

- Step 1: Check balance (add a dummy if not). Step 2: Find an initial basic feasible solution (IBFS).
- Step 3: Test optimality (MODI / u-v method). Step 4: Improve using a loop until all opportunity costs ≥ 0.

2. Initial Basic Feasible Solution (IBFS)

(a) North-West Corner Rule (NWCR)

- Start at the top-left cell; allocate min(supply, demand); cross out the exhausted row / column; move right or down.
- Fast but ignores costs – usually far from optimal.

Example:

Apply NWC to the running example.

From / To	D1	D2	D3	D4	Supply
S1	5	2	-	-	7
S2	-	6	3	-	9
S3	-	-	4	14	18
Demand	5	8	7	14	34

- Cost = 5×19 + 2×30 + 6×30 + 3×40 + 4×70 + 14×20 = 95 + 60 + 180 + 120 + 280 + 280 = **Rs 1015**.

(b) Least Cost Method (Matrix Minima)

- Allocate as much as possible in the cell with the smallest cost; cross out the satisfied row / column; repeat on the remaining cells.

> **Example:**
> Smallest cost is 8 (S3, D2): allocate 8. Next 10 (S1, D4): 7. Next 20 (S3, D4): 7. Then 40: (S2, D3): 7, (S3, D1): 3, (S2, D1): 2.

- Allocations: 8×8 + 7×10 + 7×20 + 7×40 + 3×40 + 2×70 = 64 + 70 + 140 + 280 + 120 + 140 = **Rs 814** (6 allocations ✓).

(c) Vogel's Approximation Method (VAM) – best IBFS

- Step 1: For every row and column find the **penalty** = difference between the two smallest costs.
- Step 2: Select the row / column with the largest penalty; allocate maximum in its lowest cost cell.
- Step 3: Cross out the satisfied row / column, recompute penalties, repeat. (Ties: pick the lowest cost cell.)

Iteration	Largest penalty	Allocation made
1	Column D2 (22)	S3–D2 = 8
2	Column D1 (21)	S1–D1 = 5
3	Row S3 (50)	S3–D4 = 10
4	Column D4 (50)	S1–D4 = 2
5	remaining	S2–D4 = 2, S2–D3 = 7

- Row penalties (initial): S1 = 9, S2 = 10, S3 = 12. Column penalties: D1 = 21, D2 = 22, D3 = 10, D4 = 10.
- Cost = 8×8 + 5×19 + 10×20 + 2×10 + 2×60 + 7×40 = 64 + 95 + 200 + 20 + 120 + 280 = **Rs 779**.

Method	Cost	Quality
North-West Corner	1015	worst (ignores cost)
Least Cost	814	better
VAM	779	closest to optimal

3. Moving Towards Optimality - MODI (u-v) Method

- Requires m + n - 1 occupied cells. For occupied cells: $u_i + v_j = c_{ij}$ (set u1 = 0 to start).
- For each empty cell compute the opportunity cost (net evaluation): $\Delta_{ij} = c_{ij} - (u_i + v_j)$.
- If all $\Delta_{ij} \geq 0 \rightarrow$ optimal. If some $\Delta_{ij} < 0 \rightarrow$ improve: choose the most negative cell.

Improvement by closed loop

- Draw a closed loop (horizontal / vertical moves) from the entering cell through occupied cells; mark +, -, +, - alternately.
- θ = smallest allocation on the minus cells; add θ to plus cells, subtract from minus cells. Re-test.

Example:

Test the VAM solution (cost 779) for optimality.

- Occupied: (S1,D1)=19, (S1,D4)=10, (S2,D3)=40, (S2,D4)=60, (S3,D2)=8, (S3,D4)=20.
- u1 = 0 ⇒ v1 = 19, v4 = 10; u2 = 60 - 10 = 50 ⇒ v3 = 40 - 50 = -10; u3 = 20 - 10 = 10 ⇒ v2 = 8 - 10 = -2.

Empty cell	c	u + v	Δ = c - (u + v)
S1-D2	30	-2	32
S1-D3	50	-10	60

S2-D1	₹0	69	1
S2-D2	30	48	**-18**
S3-D1	40	29	11
S3-D3	₹0	0	₹0

- Only S2-D2 is negative (-18) ⇒ not optimal. Loop: S2-D2 (+), S2-D4 (-), S3-D4 (+), S3-D2 (-).
- Minus cells hold 2 (S2-D4) and 8 (S3-D2) ⇒ θ = 2. S2-D2 = 2, S2-D4 = 0, S3-D4 = 12, S3-D2 = 6.

From / To	D1	D2	D3	D4	Supply
S1	5	-	-	2	₹
S2	-	2	₹	-	9
S3	-	6	-	12	18
Demand	5	8	₹	14	34

- New cost = 5×19 + 2×10 + 2×30 + ₹×40 + 6×8 + 12×20 = 95 + 20 + 60 + 280 + 48 + 240 = **743** (= ₹79 - 18 × 2 ✓).
- Re-test: u1 = 0, v1 = 19, v4 = 10, u3 = 10, v2 = -2, u2 = 32, v3 = 8. All Δ for empty cells: 32, 42, 19, 18, 11, 52 ≥ 0 ⇒ **optimal, Min cost = Rs 743**.

> **Note:**
> Cost change after a loop = θ × Δ (here 2 × -18 = -36). Zero Δ for an empty cell ⇒ alternative optimal solution.

4. Unbalanced, Maximisation and Degeneracy

Unbalanced transportation problem

- If supply > demand: add a **dummy destination** with demand = excess supply, all costs 0.

- If demand > supply: add a ***dummy source*** with supply = shortage, all costs 0.
- Solve normally; shipments to a dummy mean unused capacity / unmet demand.

Example:

Supply 20, 30, 25 (total 75); demand 25, 30, 10 (total 65). Excess supply 10.

From / To	D1	D2	D3	Dummy	Supply
S1	4	6	8	0	20
S2	5	3	7	0	30
S3	6	4	5	0	25
Demand	25	30	10	10	75

- Now balanced (75 = 75); solve by VAM / MODI as usual.

Maximisation problem

- Objective is profit: convert to minimisation by subtracting every entry from the largest entry (opportunity loss matrix), or multiply the matrix by -1.
- Solve as a minimisation; the optimum profit = Σ (original profit $\times$ x_{ij}).

Example:

Profit matrix: largest element 20 → new cost = 20 – profit, then solve by VAM and MODI.

Degeneracy in transportation

Definition:

A basic feasible solution is degenerate if the number of occupied cells is less than m + n – 1.

- Occurs when a row and a column are satisfied at the same time during allocation.

- Remedy: allocate a very small quantity ε (ε → 0) to an empty cell such that no closed loop is formed (an independent position) and treat it as occupied.
- During optimality, ε may also appear when a minus cell ties for θ: keep one as ε.
- At the end put ε = 0 in the final answer.

Note:
Check the number of allocations = m + n - 1 before every MODI test.

5. Assignment Problem - Introduction

Definition:
The assignment problem assigns n jobs to n workers (machines) one-to-one so that total cost (time) is minimum or total profit is maximum.

- It is a special transportation problem with all supplies = demands = 1.

Mathematical formulation

Formula:
Min $Z = \sum \sum c_{ij} x_{ij}$, x_{ij} = 1 if worker i is assigned to job j, 0 otherwise
$\sum_j x_{ij} = 1$ (each worker one job), $\sum_i x_{ij} = 1$ (each job one worker)

- Solved by the ***Hungarian method*** (Kuhn, 1955; based on KÖnig's theorem).
- Key property: subtracting a constant from any row or column does not change the optimal assignment.

6. Hungarian Algorithm

- Step 1: Make the matrix square (add dummy row / column with zeros if unbalanced).

- Step 2: ***Row reduction*** – subtract the smallest element of each row from that row.
- Step 3: ***Column reduction*** – subtract the smallest element of each column from that column.
- Step 4: Cover all zeros with the ***minimum number of horizontal and vertical lines***.
- Step 5: If number of lines = n → optimal assignment possible (go to step 7).
- Step 6: If lines < n: find the smallest uncovered element k; subtract k from every uncovered element, add k to elements at line intersections; return to step 4.
- Step 7: Assign at zeros: start with rows / columns having a single zero, cross out the row and column, continue.

Example:

Assign 4 jobs to 4 men to minimise cost.

Man / Job	J1	J2	J3	J4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

- Row minima 10, 5, 11, 8 ⇒

Man	J1	J2	J3	J4
A	0	2	9	1
B	0	5	2	3
C	1	3	2	0
D	0	7	3	1

- Column minima 0, 2, 2, 0 ⇒

Man	J1	J2	J3	J4
A	0	0	~~7~~	1
B	0	3	0	3
C	1	1	0	0
D	0	5	1	1

- Four lines needed (n = 4) ⇒ assignment: A → J2, B → J3, C → J4, D → J1 (unique zeros).
- Cost = 12 + ~~7~~ + 11 + 8 = **38**.

7. Hungarian Method – Case with Fewer Lines

Example:

Minimise the cost.

Man / Job	J1	J2	J3	J4
A	1	4	6	3
B	9	~~7~~	10	9
C	4	5	11	~~7~~
D	8	~~7~~	8	5

- Row minima 1, ~~7~~, 4, 5 ⇒ rows: A (0 3 5 2), B (2 0 3 2), C (0 1 ~~7~~ 3), D (3 2 3 0).
- Column minima 0, 0, 3, 0 ⇒ subtract 3 from column 3.

Man	J1	J2	J3	J4
A	0	3	2	2
B	2	0	0	2
C	0	1	4	3
D	3	2	0	0

- Zeros can be covered by 3 lines (column J1, row B, row D) < 4 ⇒ improve.
- Smallest uncovered element = 1 (cell C–J2). Subtract 1 from uncovered, add 1 at intersections (B–J1, D–J1).

Man	J1	J2	J3	J4
A	0	2	1	1
B	3	0	0	2
C	0	0	3	2
D	4	2	0	0

- Now 4 lines needed. A has a single zero ⇒ A → J1; C → J2; B → J3; D → J4.
- Cost = 1 + 5 + 10 + 5 = **21**.

> **Note:**
> If at the end a row (or column) has more than one zero in all choices, more than one optimal assignment exists (multiple solutions) – all have the same cost.

8. Multiple Solutions, Unbalanced and Maximisation

Multiple optimal solutions

- When, after reduction, some rows / columns have two zeros and the assignment can be made in different ways → alternative solutions with equal cost. Write each.

> **Example:**
> Reduced matrix rows (0 0 3), (0 2 0), (3 0 0) ⇒ A→1, B→3, C→2 or A→2, B→1, C→3: two optimal assignments.

Unbalanced assignment problem

- Number of workers ≠ number of jobs: add dummy jobs / workers with zero cost to get a square matrix.
- A worker assigned to a dummy job stays idle; a job assigned to a dummy worker is not done.

Example:

3 workers, 4 jobs: add a dummy worker row (0, 0, 0, 0) and apply Hungarian method to the 4 × 4 matrix.

Maximisation (profit) assignment

- Either multiply the matrix by −1 and add a constant to remove negatives, or subtract every element from the largest element (regret matrix); then minimise.

Example:

Profit matrix for 3 salesmen and 3 areas:

Salesman	Area 1	Area 2	Area 3
P	12	10	8
Q	9	11	7
R	10	6	13

- Largest = 13 ⇒ regret (13 − profit):

Salesman	Area 1	Area 2	Area 3
P	1	3	5
Q	4	2	6
R	3	7	0

- Row minima 1, 2, 0 ⇒ (0 2 4), (2 0 4), (3 7 0); column minima are 0 ⇒ assignment: P → 1, Q → 2, R → 3.
- Maximum profit = 12 + 11 + 13 = **36**.

Restricted assignment

- If a worker cannot do a job, put a very large cost M in that cell.

9. Practice Problems with Answers

- 1) Find NWC cost: supply 20, 30; demand 10, 25, 15; costs row1 (2, 3, 1), row2 (5, 4, 8). Ans: x11 = 10, x12 = 10, x22 = 15, x23 = 15; cost = 20 + 30 + 60 + 120 = 230.
- 2) How many occupied cells are required for a 4 × 5 transportation table? Ans: 4 + 5 - 1 = 8.
- 3) What is done if supply exceeds demand? Ans: add a dummy destination with zero costs.
- 4) Solve the assignment (min): rows (9, 11, 14), (6, 15, 13), (12, 13, 6). Ans: row minima 9, 6, 6 ⇒ (0 2 5), (0 9 7), (6 7 0); col min 0, 2, 0 ⇒ (0 0 5), (0 7 7), (6 5 0): 1→2, 2→1, 3→3 ⇒ cost = 11 + 6 + 6 = 23.
- 5) Differentiate the transportation and assignment problems. Ans: assignment has square matrix, supply = demand = 1, $x \in \{0, 1\}$.
- 6) What is an ε allocation? Ans: a tiny quantity placed to handle degeneracy.

> **Note:**
> Quick check: total of allocations in each row / column must equal supply / demand.

Quick Revision - Unit II

- Balanced: Σ supply = Σ demand; allocations = m + n - 1.
- IBFS: NWC (no cost), Least cost, VAM (penalty = two smallest difference).
- MODI: $u_i + v_j = c_{ij}$ for occupied; $\Delta = c - (u + v) \geq 0$ for all empty ⇒ optimal.
- Assignment: row reduce, column reduce, cover zeros, lines = n ⇒ assign.
- Dummy for unbalanced; max → subtract from largest.